

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium (LABORATORY PERFORMANCE)

COURSE CODE:

CSA09 COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO4: Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 1

Total Marks: 50

ANNEXURE A

LABORATORY PERFORMANCE

Code of Conduct:

I Challa Priyadarshini(192311118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ch Priyadarshini

ANNEXURE

LABORATORY PERFORMANCE

1. Student Registration Applet

Develop a Java Applet for a university student registration form. The applet should display labels and input controls for Student Name, Register Number, Department, Gender, and Email ID. The application should use appropriate AWT controls such as Label, TextField, Checkbox, CheckboxGroup, and Button. Use a suitable layout manager to organize the controls. When the student clicks the Submit button, the entered information should be displayed within the applet window.

The student must demonstrate:

1. Applet initialization and architecture.
2. AWT controls.
3. Layout management.
4. Button event handling.
5. Appropriate event listener interface.
6. Display of user-entered information.

2. Simple Calculator Applet

Develop an Applet-based calculator that accepts two numbers and provides buttons for Addition, Subtraction, Multiplication, and Division.

The application should use TextField, Label, and Button controls. Appropriate event listeners must be implemented so that clicking each button performs the corresponding operation.

The program must handle invalid numeric input and division by zero appropriately.

The student must demonstrate:

1. Basic Applet structure.
2. AWT controls.
3. Event sources and event objects.
4. ActionListener.
5. Exception handling.
6. Display methods.

3. Course Selection Applet

Develop a university course selection Applet where students can select:

Department

Semester

Course

Elective type

Use Choice, List, Checkbox, and Button controls wherever appropriate.

When the student clicks Register, display the selected courses and other information.

Use appropriate layout managers and event listeners.

4. Traffic Signal Control Applet

Develop an AWT Applet that simulates a traffic signal with three options:

Red

Yellow

Green

Use appropriate controls to allow the user to select a signal. When the user selects a signal, display the corresponding message such as STOP, WAIT, or GO.

Use appropriate event sources, event classes, and listener interfaces.

Question 1:

1. Student Registration Applet

Develop a Java Applet for a university student registration form. The applet should display labels and input controls for Student Name, Register Number, Department, Gender, and Email ID. The application should use appropriate AWT controls such as Label, TextField, Checkbox, CheckboxGroup, and Button. Use a suitable layout manager to organize the controls. When the student clicks the Submit button, the entered information should be displayed within the applet window.

The student must demonstrate:

1. Applet initialization and architecture.
2. AWT controls.
3. Layout management.
4. Button event handling.
5. Appropriate event listener interface.
6. Display of user-entered information.

Program:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class StudentRegistration extends Applet implements ActionListener {
    Label nameLabel, regLabel, deptLabel, genderLabel, emailLabel;
    TextField nameField, regField, emailField;
    Choice deptChoice;
    Checkbox male, female;
    CheckboxGroup genderGroup;
    Button submitButton;
    TextArea resultArea;

    public void init() {
        setBackground(new Color(230, 240, 255));
        Font normalFont = new Font("Arial", Font.PLAIN, 18);
```

```
Font labelFont = new Font("Arial", Font.BOLD, 18);
Font buttonFont = new Font("Arial", Font.BOLD, 18);
setLayout(new GridLayout(7, 2, 10, 10));
nameLabel = new Label("Student Name:");
nameLabel.setFont(labelFont);
nameField = new TextField();
nameField.setFont(normalFont);
regLabel = new Label("Register Number:");
regLabel.setFont(labelFont);
regField = new TextField();
regField.setFont(normalFont);
deptLabel = new Label("Department:");
deptLabel.setFont(labelFont);
deptChoice = new Choice();
deptChoice.setFont(normalFont);
deptChoice.add("CSE");
deptChoice.add("ECE");
deptChoice.add("EEE");
deptChoice.add("IT");
deptChoice.add("MECH");
genderLabel = new Label("Gender:");
genderLabel.setFont(labelFont);
genderGroup = new CheckboxGroup();
male = new Checkbox("Male", genderGroup, false);
male.setFont(normalFont);
female = new Checkbox("Female", genderGroup, false);
female.setFont(normalFont);
Panel genderPanel = new Panel();
genderPanel.setBackground(new Color(230, 240, 255));
genderPanel.add(male);
```

```
genderPanel.add(female);

emailLabel = new Label("Email ID:");
emailLabel.setFont(labelFont);
emailField = new TextField();
emailField.setFont(normalFont);
submitButton = new Button("Submit");
submitButton.setFont(buttonFont);
submitButton.setBackground(new Color(50, 100, 200));
submitButton.setForeground(Color.WHITE);
submitButton.addActionListener(this);
resultArea = new TextArea();
resultArea.setFont(normalFont);
resultArea.setEditable(false);
add(nameLabel);
add(nameField);
add(regLabel);
add(regField);
add(deptLabel);
add(deptChoice);
add(genderLabel);
add(genderPanel);
add(emailLabel);
add(emailField);
add(submitButton);
add(new Label(""));
Label resultLabel = new Label("Registration Details:");
resultLabel.setFont(labelFont);
add(resultLabel);
add(resultArea);
}
```

```

public void actionPerformed(ActionEvent e) {

    String name = nameField.getText();

    String department = deptChoice.getSelectedItem();

    String gender;

    } else {

        gender = "Not Selected";

    }

    String email = emailField.getText();

    resultArea.setText(

        "Student Registration Details\n\n" +

        "Student Name: " + name + "\n" +

        "Register Number: " + regNo + "\n" +

        "Department: " + department + "\n" +

        "Gender: " + gender + "\n" +

        "Email ID: " + email

    );

}

}

```

Output:

Applet Viewer: StudentRegistration.class

Applet

Student Name: Challa Priyadarshini

Register Number: 192311118

Department: CSE

Gender: ☒ Male ☐ Female

Email ID: priyadarshini72@gmail.com

Submit

Registration Details:

Department: CSE
Gender: Female
Email ID: priyadarshini72@gmail.com

Applet started.

Question 2:

Simple Calculator Applet

Develop an Applet-based calculator that accepts two numbers and provides buttons for Addition, Subtraction, Multiplication, and Division. The application should use TextField, Label, and Button controls. Appropriate event listeners must be implemented so that clicking each button performs the corresponding operation. The program must handle invalid numeric input and division by zero appropriately.

The student must demonstrate:

1. Basic Applet structure.
2. AWT controls.
3. Event sources and event objects.
4. ActionListener.
5. Exception handling.
6. Display methods.

Program:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class SimpleCalculator extends Applet implements ActionListener {

    Label titleLabel;

    Label firstNumberLabel;

    Label secondNumberLabel;

    Label resultLabel;

    TextField firstNumberField;

    TextField secondNumberField;

    TextField resultField;

    Button addButton;

    Button subtractButton;

    Button multiplyButton;

    Button divideButton;
```



```
Button clearButton;

public void init() {

    setBackground(new Color(225, 240, 255));

    setForeground(new Color(30, 30, 30));

    Font titleFont = new Font("Arial", Font.BOLD, 24);
    Font labelFont = new Font("Arial", Font.BOLD, 18);
    Font inputFont = new Font("Arial", Font.PLAIN, 18);
    Font buttonFont = new Font("Arial", Font.BOLD, 16);
    Font resultFont = new Font("Arial", Font.BOLD, 18);

    setLayout(new BorderLayout(10, 10));

    titleLabel = new Label("SIMPLE CALCULATOR", Label.CENTER);
    titleLabel.setFont(titleFont);
    titleLabel.setForeground(new Color(25, 70, 130));
    titleLabel.setBackground(new Color(180, 215, 250));
    add(titleLabel, BorderLayout.NORTH);

    Panel inputPanel = new Panel(new GridLayout(3, 2, 12, 12));
    inputPanel.setBackground(new Color(225, 240, 255));

    firstNumberLabel = new Label("First Number:");
    firstNumberLabel.setFont(labelFont);
    firstNumberLabel.setForeground(new Color(25, 60, 110));
    firstNumberField = new TextField();
    firstNumberField.setFont(inputFont);
    firstNumberField.setBackground(Color.WHITE);
    firstNumberField.setForeground(Color.BLACK);

    secondNumberLabel = new Label("Second Number:");
    secondNumberLabel.setFont(labelFont);
    secondNumberLabel.setForeground(new Color(25, 60, 110));
    secondNumberField = new TextField();
    secondNumberField.setFont(inputFont);
    secondNumberField.setBackground(Color.WHITE);
```

```
secondNumberField.setForeground(Color.BLACK);
resultLabel = new Label("Result:");
resultLabel.setFont(labelFont);
resultLabel.setForeground(new Color(25, 60, 110));
resultField = new TextField();
resultField.setFont(resultFont);
resultField.setBackground(new Color(245, 250, 255));
resultField.setForeground(new Color(0, 90, 40));
resultField.setEditable(false);
inputPanel.add(firstNumberLabel);
inputPanel.add(firstNumberField);
inputPanel.add(secondNumberLabel);
inputPanel.add(secondNumberField);
inputPanel.add(resultLabel);
inputPanel.add(resultField);
add(inputPanel, BorderLayout.CENTER);
Panel buttonPanel = new Panel(new GridLayout(1, 5, 10, 10));
buttonPanel.setBackground(new Color(225, 240, 255));
addButton = new Button("ADD");
subtractButton = new Button("SUBTRACT");
multiplyButton = new Button("MULTIPLY");
divideButton = new Button("DIVIDE");
clearButton = new Button("CLEAR");
addButton.setFont(buttonFont);
subtractButton.setFont(buttonFont);
multiplyButton.setFont(buttonFont);
divideButton.setFont(buttonFont);
clearButton.setFont(buttonFont);
addButton.setBackground(new Color(60, 150, 80));
addButton.setForeground(Color.WHITE);
```

```

subtractButton.setBackground(new Color(220, 150, 40));
subtractButton.setForeground(Color.WHITE);
multiplyButton.setBackground(new Color(120, 80, 180));
multiplyButton.setForeground(Color.WHITE);
divideButton.setBackground(new Color(50, 110, 190));
divideButton.setForeground(Color.WHITE);
clearButton.setBackground(new Color(190, 60, 60));
clearButton.setForeground(Color.WHITE);
addButton.addActionListener(this);
subtractButton.addActionListener(this);
multiplyButton.addActionListener(this);
divideButton.addActionListener(this);
clearButton.addActionListener(this);
buttonPanel.add(addButton);
buttonPanel.add(subtractButton);
buttonPanel.add(multiplyButton);
buttonPanel.add(divideButton);
buttonPanel.add(clearButton);
add(buttonPanel, BorderLayout.SOUTH);
}

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == clearButton) {
        firstNumberField.setText("");
        secondNumberField.setText("");
        resultField.setText("");
        return;
    }
    try {
        double number1 =
            Double.parseDouble(firstNumberField.getText().trim());

```

```

double number2 =

    Double.parseDouble(secondNumberField.getText().trim());
double result = 0;
if (e.getSource() == addButton) {
    result = number1 + number2;
    resultField.setText(String.valueOf(result));
}
else if (e.getSource() == subtractButton) {
    result = number1 - number2;
    resultField.setText(String.valueOf(result));
}
else if (e.getSource() == multiplyButton) {
    result = number1 * number2;
    resultField.setText(String.valueOf(result));
}
else if (e.getSource() == divideButton) {
    if (number2 == 0) {
        resultField.setText("Cannot divide by zero");
    } else {
        result = number1 / number2;
        resultField.setText(String.valueOf(result));
    }
}

}

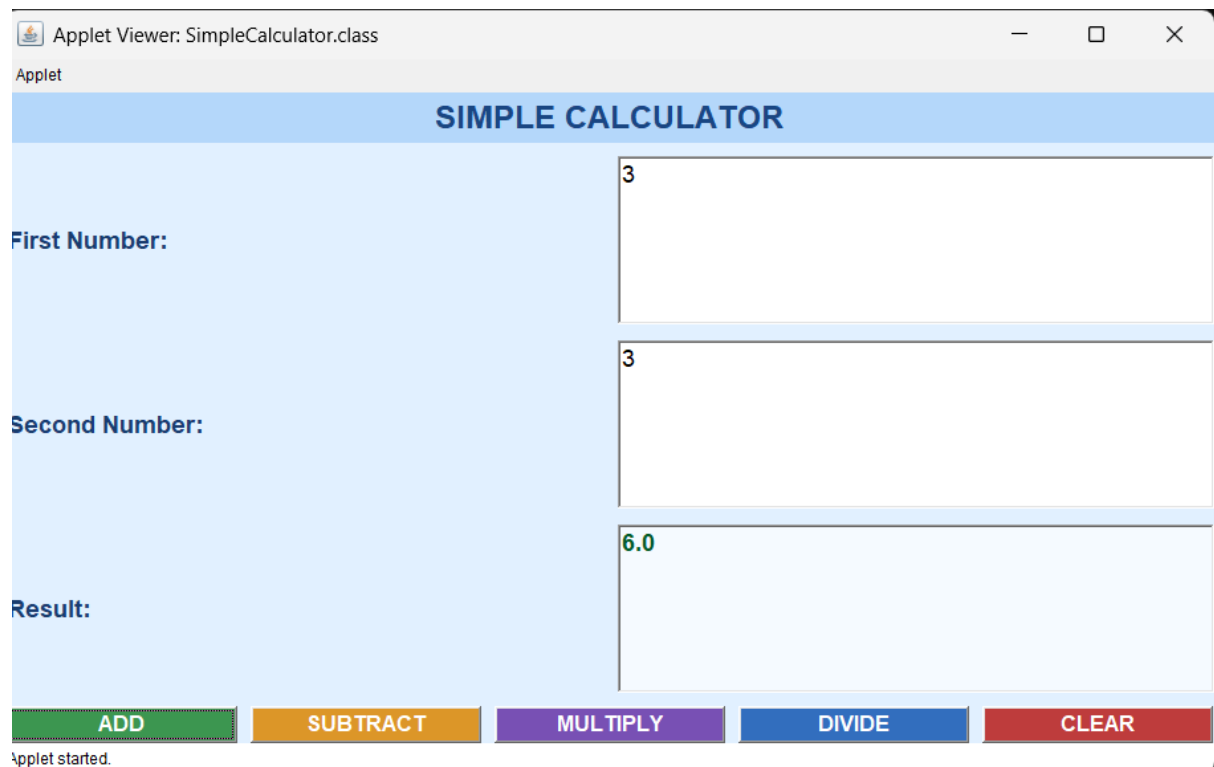
catch (NumberFormatException ex) {
    resultField.setText("Invalid number!");
}

}

}

```

Output:



Question 3:

Course Selection Applet

Develop a university course selection Applet where students can select:

Department

Semester

Course

Elective type

Use Choice, List, Checkbox, and Button controls wherever appropriate.

When the student clicks Register, display the selected courses and other information.

Use appropriate layout managers and event listeners.

Program:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class CourseSelection extends Applet implements ActionListener {
    Label titleLabel;
```

```
Label departmentLabel;
Label semesterLabel;
Label courseLabel;
Label electiveLabel;
Label resultLabel;
Choice departmentChoice;
Choice semesterChoice;
List courseList;
Checkbox coreCheckbox;
Checkbox electiveCheckbox;
CheckboxGroup electiveGroup;
Button registerButton;
Button clearButton;
TextArea resultArea;
public void init() {
    setBackground(new Color(230, 242, 255));
    setForeground(new Color(30, 30, 30));
    Font titleFont =
        new Font("Arial", Font.BOLD, 24);
    Font labelFont =
        new Font("Arial", Font.BOLD, 18);
    Font normalFont =
        new Font("Arial", Font.PLAIN, 17);
    Font buttonFont =
        new Font("Arial", Font.BOLD, 16);
    setLayout(new BorderLayout(10, 10));
    titleLabel =
        new Label("UNIVERSITY COURSE SELECTION", Label.CENTER);
    titleLabel.setFont(titleFont);
    titleLabel.setForeground(new Color(20, 70, 130));
```

```
titleLabel.setBackground(new Color(180, 215, 250));
add(titleLabel, BorderLayout.NORTH);
Panel mainPanel = new Panel(new GridLayout(4, 2, 12, 12));
mainPanel.setBackground(new Color(230, 242, 255));
departmentLabel =
    new Label("Department:");
departmentLabel.setFont(labelFont);
departmentLabel.setForeground(
    new Color(25, 60, 110)
);
departmentChoice = new Choice();
departmentChoice.setFont(normalFont);
departmentChoice.add("Computer Science and Engineering");
departmentChoice.add("Electronics and Communication");
departmentChoice.add("Electrical and Electronics");
departmentChoice.add("Information Technology");
departmentChoice.add("Mechanical Engineering");
semesterLabel =
    new Label("Semester:");
semesterLabel.setFont(labelFont);
semesterLabel.setForeground(
    new Color(25, 60, 110)
);
semesterChoice = new Choice();
semesterChoice.setFont(normalFont);
semesterChoice.add("Semester 1");
semesterChoice.add("Semester 2");
semesterChoice.add("Semester 3");
semesterChoice.add("Semester 4");
semesterChoice.add("Semester 5");
```

```
semesterChoice.add("Semester 6");
semesterChoice.add("Semester 7");
semesterChoice.add("Semester 8");
mainPanel.add(departmentLabel);
mainPanel.add(departmentChoice);
mainPanel.add(semesterLabel);
mainPanel.add(semesterChoice);
courseLabel =
    new Label("Select Course(s):");
courseLabel.setFont(labelFont);
courseLabel.setForeground(
    new Color(25, 60, 110)
);
courseList =
    new List(5, true);
courseList.setFont(normalFont);
courseList.add("Data Structures");
courseList.add("Database Management System");
courseList.add("Computer Networks");
courseList.add("Operating Systems");
courseList.add("Web Development");
courseList.add("Artificial Intelligence");
courseList.add("Machine Learning");
courseList.add("Cloud Computing");
mainPanel.add(courseLabel);
mainPanel.add(courseList);
electiveLabel =
    new Label("Elective Type:");
electiveLabel.setFont(labelFont);
electiveLabel.setForeground(
```



```

        new Color(25, 60, 110)
    );
    electiveGroup =
        new CheckboxGroup();
    coreCheckbox =
        new Checkbox(
            "Core",
            electiveGroup,
            true
        );
    electiveCheckbox =
        new Checkbox(
            "Elective",
            electiveGroup,
            false
        );
    coreCheckbox.setFont(normalFont);
    electiveCheckbox.setFont(normalFont);
    Panel electivePanel =
        new Panel(new FlowLayout(FlowLayout.LEFT));
    electivePanel.setBackground(
        new Color(230, 242, 255)
    );
    electivePanel.add(coreCheckbox);
    electivePanel.add(electiveCheckbox);
    mainPanel.add(electiveLabel);
    mainPanel.add(electivePanel);
    add(mainPanel, BorderLayout.CENTER);
    Panel buttonPanel =
        new Panel(new FlowLayout(FlowLayout.CENTER, 15, 10));

```

```
buttonPanel.setBackground(  
    new Color(230, 242, 255)  
);  
registerButton =  
    new Button("REGISTER");  
clearButton =  
    new Button("CLEAR");  
registerButton.setFont(buttonFont);  
clearButton.setFont(buttonFont);  
registerButton.setBackground(  
    new Color(45, 140, 75)  
);  
registerButton.setForeground(Color.WHITE);  
clearButton.setBackground(  
    new Color(190, 60, 60)  
);  
clearButton.setForeground(Color.WHITE);  
registerButton.addActionListener(this);  
clearButton.addActionListener(this);  
buttonPanel.add(registerButton);  
buttonPanel.add(clearButton);  
Panel bottomPanel =  
    new Panel(new BorderLayout(5, 5));  
bottomPanel.setBackground(  
    new Color(230, 242, 255)  
);  
resultLabel =  
    new Label(  
        "Registration Details",  
        Label.CENTER
```

```
);  
resultLabel.setFont(labelFont);  
resultLabel.setForeground(  
    new Color(20, 70, 130)  
);  
resultArea =  
    new TextArea(6, 40);  
resultArea.setFont(normalFont);  
resultArea.setBackground(  
    new Color(250, 252, 255)  
);  
resultArea.setForeground(  
    new Color(20, 70, 40)  
);  
resultArea.setEditable(false);  
bottomPanel.add(  
    resultLabel,  
    BorderLayout.NORTH  
);  
bottomPanel.add(  
    resultArea,  
    BorderLayout.CENTER  
);  
Panel southPanel =  
    new Panel(new BorderLayout());  
southPanel.setBackground(  
    new Color(230, 242, 255)  
);  
southPanel.add(  
    buttonPanel,
```

```

        BorderLayout.NORTH
    );
    southPanel.add(
        bottomPanel,
        BorderLayout.CENTER
    );
    add(
        southPanel,
        BorderLayout.SOUTH
    );
}

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == clearButton) {
        courseList.deselect(
            0
        );
        for (int i = 0;
            i < courseList.getItemCount();
            i++) {

            courseList.deselect(i);
        }
        departmentChoice.select(0);
        semesterChoice.select(0);
        electiveGroup.setSelectedCheckbox(
            coreCheckbox
        );
        resultArea.setText("");
        return;
    }
}

```

```

if (e.getSource() == registerButton) {
    String department =
        departmentChoice.getSelectedItemAt();
    String semester =
        semesterChoice.getSelectedItemAt();
    String electiveType =
        electiveGroup
        .getSelectedCheckbox()
        .getLabel();
    String[] selectedCourses =
        courseList.getSelectedItems();
    if (selectedCourses.length == 0) {
        resultArea.setText(
            "Please select at least one course."
        );
        return;
    }
    String result =
        "COURSE REGISTRATION SUCCESSFUL\n\n";
    result +=
        "Department : "
        + department
        + "\n";
    result +=
        "Semester  : "
        + semester
        + "\n";
    result +=
        "Elective Type : "
        + electiveType

```

```
        + "\n\n";
result +=
    "Selected Courses:\n";
for (int i = 0;
    i < selectedCourses.length;
    i++) {
result +=
    (i + 1)
    + ". "
    + selectedCourses[i]
    + "\n";
}
resultArea.setText(result);
}
}
}
```

Output:

Question 4.

Traffic Signal Control Applet

Develop an AWT Applet that simulates a traffic signal with three options:

Red

Yellow

Green

Use appropriate controls to allow the user to select a signal. When the user selects a signal, display the corresponding message such as STOP, WAIT, or GO.

Use appropriate event sources, event classes, and listener interfaces.

Program:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class TrafficSignal extends Applet implements ActionListener {
```

```
Label titleLabel;
Label instructionLabel;
Label messageLabel;
Choice signalChoice;
Button selectButton;
String selectedSignal = "Red";
String message = "STOP";
public void init() {
    setBackground(new Color(225, 240, 255));
    setForeground(Color.BLACK);
    Font titleFont =
        new Font("Arial", Font.BOLD, 26);
    Font labelFont =
        new Font("Arial", Font.BOLD, 18);
    Font normalFont =
        new Font("Arial", Font.PLAIN, 18);
    Font buttonFont =
        new Font("Arial", Font.BOLD, 17);
    setLayout(new BorderLayout(10, 10));
    titleLabel =
        new Label(
            "TRAFFIC SIGNAL CONTROL",
            Label.CENTER
        );
    titleLabel.setFont(titleFont);
    titleLabel.setForeground(
        new Color(20, 60, 120)
    );
    titleLabel.setBackground(
        new Color(180, 215, 250)
    );
}
```



```
);  
add(titleLabel, BorderLayout.NORTH);  
Panel controlPanel =  
    new Panel(  
        new FlowLayout(  
            FlowLayout.CENTER,  
            15,  
            12  
        )  
    );
```

```
controlPanel.setBackground(  
    new Color(225, 240, 255)  
);  
instructionLabel =  
    new Label("Select Signal:");  
instructionLabel.setFont(labelFont);  
instructionLabel.setForeground(  
    new Color(30, 60, 100)  
);
```

```
signalChoice = new Choice();  
signalChoice.setFont(normalFont);  
signalChoice.add("Red");  
signalChoice.add("Yellow");  
signalChoice.add("Green");  
selectButton =  
    new Button("SELECT SIGNAL");  
selectButton.setFont(buttonFont);  
selectButton.setBackground(  
    new Color(45, 100, 180)
```

```

);
selectButton.setForeground(Color.WHITE);
selectButton.addActionListener(this);
controlPanel.add(instructionLabel);
controlPanel.add(signalChoice);
controlPanel.add(selectButton);
add(controlPanel, BorderLayout.SOUTH);
messageLabel =
    new Label(
        "STOP",
        Label.CENTER
    );
messageLabel.setFont(
    new Font("Arial", Font.BOLD, 28)
);
messageLabel.setForeground(Color.RED);
add(messageLabel, BorderLayout.CENTER);
}

public void actionPerformed(ActionEvent e) {
    selectedSignal =
        signalChoice.getSelectedItem();
    if (selectedSignal.equals("Red")) {
        message = "STOP";
        messageLabel.setText(message);
        messageLabel.setForeground(Color.RED);
    }
    else if (selectedSignal.equals("Yellow")) {
        message = "WAIT";
        messageLabel.setText(message);
        messageLabel.setForeground(

```

```

        new Color(220, 150, 0)
    );
}
else if (selectedSignal.equals("Green")) {
    message = "GO";
    messageLabel.setText(message);
    messageLabel.setForeground(
        new Color(0, 150, 50)
    );
    repaint();
}

public void paint(Graphics g) {
    super.paint(g);
    int x = 250;
    int y = 100;
    int width = 120;
    int height = 300;
    g.setColor(new Color(45, 45, 45));
    g.fillRoundRect(
        x,
        y,
        width,
        height,
        20,
        20
    );
    g.setColor(Color.BLACK);
    g.drawRoundRect(
        x,
        y,

```

```
        width,
        height,
        20,
        20
    );
    if (selectedSignal.equals("Red")) {
        g.setColor(Color.RED);
    } else {
        g.setColor(new Color(90, 30, 30));
    }
    g.fillOval(
        x + 30,
        y + 20,
        60,
        60
    );
    if (selectedSignal.equals("Yellow")) {
        g.setColor(Color.YELLOW);
    } else {
        g.setColor(new Color(100, 90, 20));
    }
    g.fillOval(
        x + 30,
        y + 110,
        60,
        60
    );
    if (selectedSignal.equals("Green")) {
        g.setColor(Color.GREEN);
    } else {
```

```
        g.setColor(new Color(20, 80, 30));
    }
    g.fillOval(
        x + 30,
        y + 200,
        60,
        60
    );
    g.setFont(
        new Font("Arial", Font.BOLD, 22)
    );
    if (selectedSignal.equals("Red")) {
        g.setColor(Color.RED);
        g.drawString(
            "STOP",
            395,
            160
        );
    } else if (selectedSignal.equals("Yellow")) {
        g.setColor(
            new Color(220, 150, 0)
        );
        g.drawString(
            "WAIT",
            395,
            160
        );
    } else {
        g.setColor(
            new Color(0, 140, 50)
```

```
);  
g.drawString(  
    "GO",  
    395,  
    160  
);  
}  
}  
}
```

Output:



TRAFFIC SIGNAL CONTROL

WAIT

Select Signal:

Yellow

SELECT SIGNAL

Applet started.

TRAFFIC SIGNAL CONTROL

GO

Select Signal:

Green

SELECT SIGNAL

Applet started.

